

Christopher Zhang

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Education

Brown University, *Sc.B. Computer Science & A.B. Business Economics*

Providence, RI

- **GPA:** 4.00/4.00, Varsity Student-Athlete

Expected Graduation May 2028

- **Relevant Coursework:** SWE, Data Structures and Algorithms, AI, Deep Learning, Computer Systems, Discrete Math

Experience

Scout Motors

Charlotte, NC

AI Engineer

May 2026 - Present

- Launched an AI security assistant with Copilot Studio and Jira, reducing projected support costs by \$800K annually
- Developing an AI developer platform generating Scout-compliant apps from natural language, cutting deployment time by 75%
- Automating infrastructure provisioning and observability using AWS EKS, Argo CD, Kargo, Crossplane, and Datadog

Mandy

Remote

Software Engineer

June 2025 - August 2025

- Built an AI-powered room redesign tool, transforming user photos into styled layouts with curated Amazon product recs
- Shipped a full-stack app for 1,000+ image uploads and layout generation using Next.js, TypeScript, IMGBB, and Vercel
- Integrated Clerk authentication to track preferences and deliver tailored product recommendations and chatbot responses

Full Stack @Brown

Providence, RI

Developer

September 2025 - Present

- Built a production Next.js platform for the Brown Brain Bee, integrating a CMS API and implementing full-text search
- Developing a full-stack publishing system for Brown Music Review with article management and automated email notifications

Waves Swim Team

Charlotte, NC

Head Coach

April 2025 - July 2025

- Led and managed training for over 150 swimmers of all age levels, designing effective practice plans to maximize performance

Projects

Dating Profile Analyzer, *React, TypeScript, Node.js, Firebase, Google Gemini API*

October 2025 - January 2026

- Built a full-stack React and Node.js application that analyzes dating profiles using image scoring and AI-generated feedback
- Designed image-ranking algorithms using blur variance, brightness normalization, and smile detection-based weighted scoring
- Implemented prompt-driven Gemini AI feedback using stored personality quiz data to generate personalized profile bio recs

Travel Planner, *React, TypeScript, Node.js, Firebase, OpenStreetMap API*

September 2025 - December 2025

- Engineered an A* pathfinding system with lazy tile loading and LRU caching for 200+ mile routing across OpenStreetMap data
- Architected backend routing and trip APIs with Firebase-backed persistence to save and reload multi-day itineraries in real-time
- Implemented GDPR-compliant, consent-aware ad targeting, integrating external ad server APIs based on destination and itinerary

AI Swim Start Coach, *React, TypeScript, OpenCV, MediaPipe*

June 2025 - June 2026

- Integrated MediaPipe and OpenCV for pose detection to extract joint coordinates and movement data from video input
- Delivered AI-powered real-time feedback, helping 100+ swimmers improve their start technique with personalized insights

Extracurricular Activities

Brown University Men's Swimming and Diving

Providence, RI

NCAA Division 1 Student-Athlete

September 2024 – Present

- Balanced 25-hour training weeks, competitions, and travel while managing a full academic workload
- Set school records in the 400 Medley and 800 Free relays; ranked top 5 all-time in 50 Free, 100 Free, 200 Free, and 100 Fly

Social Media Content Creator

Remote

Founder (@chris_swimzz)

April 2024 – Present

- Produced engaging and educational swim-related content, reaching an audience of over 28,000 followers on different platforms
- Partnered with 8+ brands, including notable names such as Arena and Nike, to create authentic promotional campaigns
- Achieved over 10 million views across content, inspiring swimmers globally and fostering a supportive online community

Skills & Interests

- **Languages & Frameworks:** Java, Python, JavaScript, TypeScript, C++, React, Node.js, Express.js
- **Tools & Technologies:** Docker, AWS EKS, PostgreSQL, PyTorch, TensorFlow, Copilot Studio, Argo CD, Crossplane, Datadog
- **Interests:** Piano (currently learning Liebesträume), Ping-Pong, Journaling, Beli, Skiing, Video Editing